

Zihao (Jason) Zhang

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PROFESSIONAL SUMMARY

M.S. candidate in Cognitive and Information Sciences with national laboratory experience training and evaluating neural network models inside an established scientific codebase. Builds reproducible ML pipelines at scale in Python, Java, and Bash/Slurm, with emphasis on leakage-safe experimental design, honest model evaluation, and source-level debugging.

EDUCATION

University of California, Merced

Cognitive and Information Sciences, M.S. | GPA: 4.0/4.0

Merced, CA

Expected Aug 2026

University of California, Davis

Cognitive Science, B.S.

Davis, CA

Jun 2025

EXPERIENCE

Affiliate Intern, Machine Learning and Metagenomics

Joint Genome Institute, Lawrence Berkeley National Laboratory

May 2026 – Jul 2026

Berkeley, CA

- Led the QuickBin neural-network adaptation workflow as the primary student contributor, building reproducible Bash/Slurm pipelines around BBTools/QuickBin, Flye, minimap2, and GradeBins on Dori HPC with FASTA/SAM/TSV validation at each stage.
- Designed and scaled synthetic long-read benchmarks from 663 to 2,013 genomes (8.29 Gbp, 62,714 contigs, N50 413,899 bp), including a 50-strain close-relative stress test at 90–95% ANI targeting the binner's hardest failure mode.
- Enforced genome-level train/validation/test splits (1,435/279/299) before generating any training vectors, preventing the pair-level leakage a post-hoc split silently introduces; verified by an automated cross-split leakage check that returned zero.
- Retrained the shipping classifier through hyperparameter sweeps over learning rate, training length, network width, and stringency modes; the retained candidate cut contamination from 1.7530 to 1.3724 on a genome-held-out split for a 0.86-point recovery cost.
- Ran the project's first external evaluation on the CAMI II PacBio benchmark and reported that the internal model ranking did not transfer, documenting the confound rather than the favorable internal result.
- Debugged BBTools/QuickBin at the Java source level, patching a private clone to restore training-vector output without touching the production tree, and traced a zero-coverage failure through FASTA/SAM identifier and CIGAR parsing contracts to recover 74,895 alignment records.

Research Assistant

Janata Lab

Jun 2024 – Jun 2025

Davis, CA

- Developed a real-time eye-blink detection module integrating browser-based HTML/JavaScript workflows with Python detection logic for behavioral data collection.
- Built experiment data pipelines with Django ORM and PyEnsemble to manage datasets and enforce data-integrity constraints.

PROJECTS

Tonal Inference Modeling

Python, PyTorch

Aug 2025 – Present

- Compared a simple recurrent network against an EMA+MLP baseline for musical key inference from MIDI, isolating chord representation rather than recurrence as the actual performance bottleneck.
- Built a six-piece, difficulty-graded MIDI evaluation corpus and ran a predeclared 54-condition sensitivity analysis of a non-neural resolver with frozen constants and negative controls, characterizing the method family's limits rather than reporting a tuned best case.

AI Algorithms and Reinforcement Learning Projects

Python | 2025 coursework, independently re-verified 2026

Jan 2025 – Aug 2026

- Implemented A* with elevation-aware admissible heuristics and a Dijkstra baseline; on synthetic terrain, optimal A* matched Dijkstra's path cost while expanding 81.25% fewer nodes; weighted A* ($w = 1.8$) cut expansions 98.57% for a 1.83% cost increase.
- Recovered a Minimax + Alpha-Beta Connect Four agent, traced two prior benchmark results to measurement artifacts rather than algorithm behavior, and added terminal detection: 93/100 on held-out seeds vs an unmodified 1001-rollout Monte Carlo baseline at fixed depth 6.
- Implemented a DQN for Pong with replay buffering, target-network updates, and epsilon-greedy exploration.

SKILLS

ML and Applied AI: PyTorch, scikit-learn, Pandas, NumPy, Neural Network Training, Hyperparameter Search, Model Evaluation, Benchmark Design

Programming: Python, Java, C/C++, SQL, R, Bash

Systems and Workflow: Linux, Git, Slurm, HPC, Django ORM, Reproducible Pipelines, Integration Testing, Object-Oriented Design